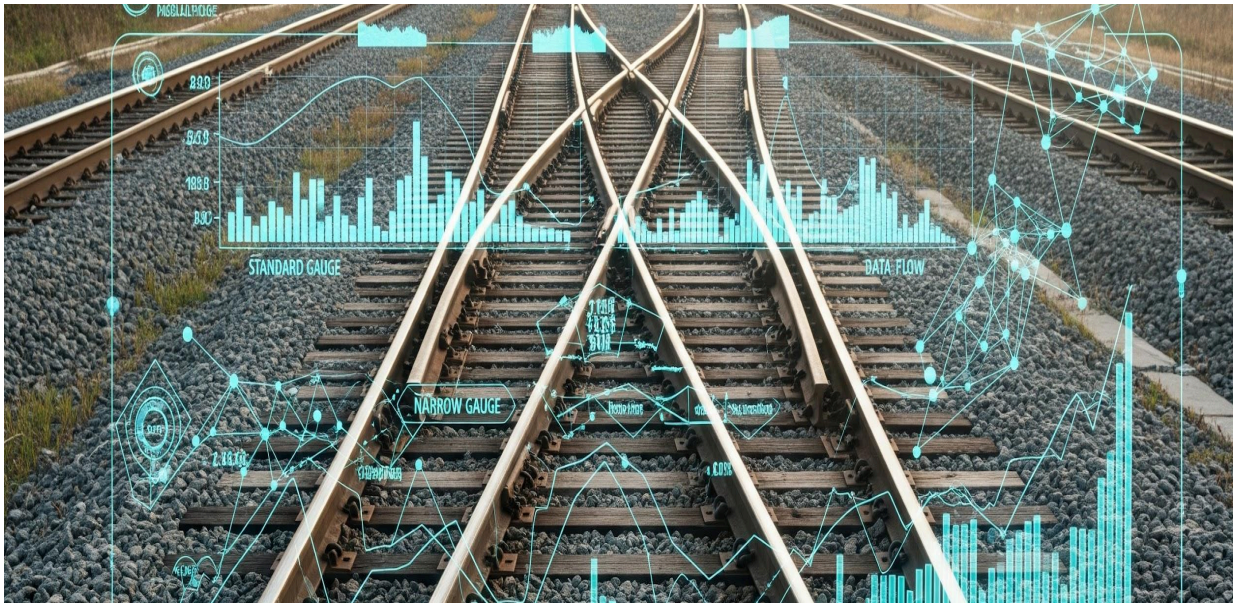




Data Analytics Project: Railway Gauge Trends

Dataset Used: railway_gauges.csv

This document outlines a structured, end-to-end data analysis project based on railway gauge data. The project is divided into sections: **Data Analytics Project: Railway Gauge Trends**

This document outlines a structured, end-to-end data analysis project based on railway gauge data.

Dataset: Railway Gauge Data (Assumed to be in a CSV file)

Project Goal: To analyze historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.

Modules Used

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (● Easy)

Description:

1. The railway gauge dataset was loaded from the CSV file using the Pandas library.
2. The dataset was stored in a Pandas DataFrame for further analysis.
3. The first five rows and column names were displayed to understand the structure of the data.
4. Missing values were checked using `isnull()` and counted using `sum()`.
5. Missing values were replaced with `0` to maintain consistency in the dataset.
6. The Broad Gauge, Metre Gauge, Narrow Gauge, and Total columns were converted into numeric data types for further calculations.

Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

Conclusion:

1. The dataset was successfully loaded into a Pandas DataFrame.
2. The initial inspection helped verify the structure and contents of the dataset.
3. The missing-value check showed 0 missing values in the provided data.

4. The gauge columns were successfully converted into numeric data types.
5. The cleaned dataset is now ready for calculations and visualization.
6. This cleaning process improves the reliability of the further analysis.

Scenario 2: Simple Visualization (● Easy)

Description:

1. The **Year** and **Total** columns were selected from the cleaned dataset.
2. A line graph was created using Matplotlib to visualize total railway tracks over the years.
3. The year was represented on the X-axis and total railway tracks on the Y-axis.
4. A suitable title and axis labels were added to make the graph understandable.
5. The year labels were rotated by 45 degrees to improve readability.
6. The graph shows fluctuations in total railway tracks across different years.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing Total tracks over the Years .
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.

Conclusion:

1. The line graph provides a clear view of changes in total railway tracks over time.
2. The total railway track length does not increase continuously and shows several fluctuations.
3. There are periods of both growth and decline in the dataset.
4. Despite these fluctuations, the overall long-term trend is increasing.
5. The visualization makes it easier to identify changes that may not be obvious from raw data.
6. Therefore, the railway network shows an overall expansion over the period analyzed.

Scenario 3: Filtering + Bar Chart (🟡 Medium)

Description:

1. The dataset was filtered to include only years after 2000.
2. Broad Gauge, Metre Gauge, and Narrow Gauge were selected for comparison.
3. A grouped bar chart was created to compare the three gauge types.
4. A separate bar was displayed for each gauge for every selected year.
5. A legend was added to identify the three gauge categories.
6. The chart clearly shows that Broad Gauge has the highest track length in the post-2000 period.

Task	Description
1.	Filter the dataset to include only years <i>after</i> 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a grouped bar chart comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Conclusion:

1. Broad Gauge is clearly the dominant gauge after 2000.
2. Metre Gauge has a considerably smaller track length compared with Broad Gauge.
3. Narrow Gauge represents the smallest portion among the three gauges in recent years.
4. The comparison shows a strong difference between Broad Gauge and the other gauges.
5. The trend suggests the increasing importance of Broad Gauge in the railway network.
6. Therefore, Broad Gauge is the major gauge in the recent period analyzed.

Scenario 4: Feature Engineering + Pie Chart (🟡 Medium)

Description:

1. The total values of Broad Gauge, Metre Gauge, and Narrow Gauge were calculated across all years.

2. The calculated totals were stored in a Pandas Series.
3. A pie chart was created to visualize the contribution of each gauge.
4. Percentage labels were added using the `autopct` parameter.
5. Broad Gauge contributes approximately 65.0% of the combined gauge totals.
6. Metre Gauge contributes 29.5%, while Narrow Gauge contributes 5.5%.

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a pie chart using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using <code>autopct</code> (e.g., '%.1f%%').
5.	Interpret which gauge contributes the most to the entire history of the railway network.

Conclusion:

1. Broad Gauge has the largest contribution among all three gauge types.
2. Its contribution of 65.0% makes it the dominant gauge in the analysis.
3. Metre Gauge has the second-largest contribution at 29.5%.
4. Narrow Gauge has the smallest contribution at 5.5%.
5. The pie chart clearly demonstrates the unequal distribution of gauge types.
6. Therefore, Broad Gauge plays the most significant role in the overall railway gauge structure.

Scenario 5: Advanced Analysis + Multiple Graphs (● Hard)

Description:

1. Percentage columns were created to calculate each gauge's contribution relative to the Total.
2. NumPy's `np.diff()` function was used to calculate year-to-year changes in total railway tracks.
3. The highest yearly growth was 125 units in 2003–04.
4. A line graph was created to compare the long-term trends of all three gauges.
5. A stacked bar chart was used to visualize the composition of the railway gauges over the years.
6. The graphs show Broad Gauge increasing while Metre Gauge and Narrow Gauge generally decline.

Task	Description
1.	Create new columns for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.
2.	Use the NumPy function np.diff to calculate the yearly growth (change) of the Total tracks column.
3.	Plot: a) A single Line Graph showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A Stacked Bar Chart showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	Highlight/Identify: a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	Final Conclusion: Based on the percentage composition line graph, provide a conclusion to the question: "Is the railway system shifting towards a single dominant gauge?"

Conclusion:

1. The analysis shows a clear long-term shift toward Broad Gauge.
2. Broad Gauge increased significantly from the beginning to the end of the dataset.
3. Metre Gauge experienced a substantial overall decline.
4. Narrow Gauge also showed an overall declining trend.
5. The highest yearly growth in total railway tracks was 125 units during 2003–04.
6. Overall, the railway system is shifting toward Broad Gauge as the dominant gauge.

Final Conclusion:

The Railway Gauge Analysis project successfully examined railway gauge data using Pandas, NumPy, and Matplotlib. The data was loaded, cleaned, transformed, and analyzed through multiple visualization techniques. The analysis identified Broad Gauge as the dominant gauge, while Metre Gauge and Narrow Gauge showed overall declining trends. The highest yearly growth in total railway tracks was 125 units in 2003–04. Overall, the results indicate a clear shift toward a railway system increasingly dominated by Broad Gauge.

What you have learned

1. I learned how to load and clean CSV data using Pandas and work with it using DataFrames.
2. I learned how to check missing values, convert data types, filter data, and perform calculations on different columns.
3. I learned how to use NumPy, especially `np.diff()`, to calculate year-to-year changes in railway track data.
4. I learned how to create different visualizations such as line graphs, grouped bar charts, pie charts, and stacked bar charts using Matplotlib.
5. I learned how to interpret graphs and identify important insights, such as Broad Gauge being the dominant gauge and the decline of Metre and Narrow Gauge.
6. I also learned how to save graphs, organize project files, and prepare a report based on the results of data analysis.